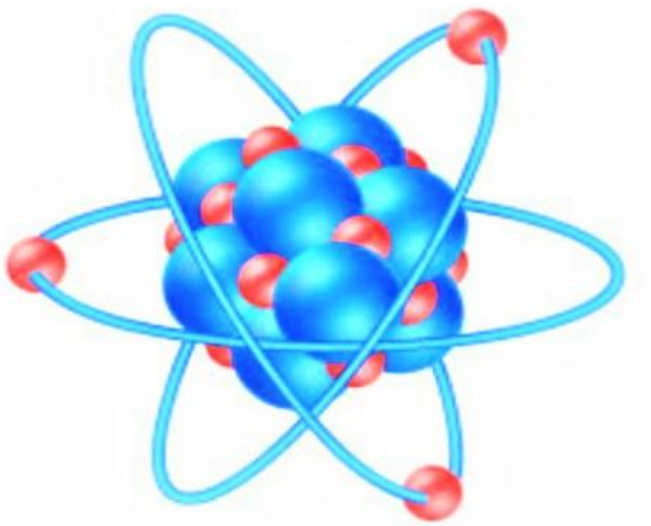
Fizika

9-sinf

Mavzu:

Masalalar yechish.



Yorug‘likning qaytish va sinish qonunlari. To‘la ichki qaytish.

$$\alpha = \beta; \quad n_{21} = \frac{\sin\alpha}{\sin\beta}; \quad n = \frac{c}{\vartheta}; \quad \frac{n_2}{n_1} = \frac{\vartheta_1}{\vartheta_2}$$

$$\frac{n_2}{n_1} = \frac{\sin\alpha}{\sin\beta}; \quad \sin\alpha_0 = \frac{n_2}{n_1}; \quad \sin\alpha_0 = \frac{1}{n}$$

Linzaning optik kuchi: $D = \frac{1}{F}; \quad F = \frac{1}{D}$

Fokus masofasi: $F = \frac{1}{(n-1)\cdot(\frac{1}{R_1}+\frac{1}{R_2})}$

$\frac{1}{F} = \frac{1}{d} + \frac{1}{f};$ $\quad -\frac{1}{F} = \frac{1}{d} - \frac{1}{f}$

1-masala

Quyosh yorug‘ligi idishdagi suv yuzuga tushmoqda. Agar sinish burchagi 42º bo‘lsa, tushish burchagini toping.

Berilgan:

$n_{21} = 1{,}33$

$\beta = 42^0$

Topish kerak: $\alpha - ?$

Formula:

$$n_{21} = \frac{sin\,\alpha}{sin\,\beta}$$

$$sin\,\alpha = n_{21} \cdot sin\,\beta$$

Yechish: $sin\,42^0 = 0{,}6691$

$$sin\,\alpha = 1{,}33 \cdot 0{,}6691 = 0{,}8899$$

$$\alpha = arksin\,0{,}8899 = 63^0$$

Javob: $\alpha = 63^0$

2-masala

Suvning sindirish koʻrsatkichi 1,33, shishaniki 1,5. Shishaning suvga nisbatan sindirish koʻrsatkichini toping.

Berilgan:

$n_1 = 1{,}33$

$n_2 = 1{,}5$

Topish kerak: $n_{21} - ?$

Formula:

$$n_{21} = \frac{n_2}{n_1}$$

Yechish: $$n_{21} = \frac{1{,}5}{1{,}33} = 1{,}13$$

Javob: $\boldsymbol{n_{21} = 1{,}13}$ **.**

3-masala

Spirt uchun to‘la qaytishning chegaraviy burchagi 47° ga teng.

Spirtning nur sindirish ko‘rsatkichini toping.

Berilgan:

$\alpha_0 = 47^0$

Topish kerak: $n-?$

Formula:

$$\sin \alpha_0 = \frac{1}{n}$$

$$n = \frac{1}{\sin \alpha_0}$$

Yechish: $n = \frac{1}{\sin 47^0} = \frac{1}{0{,}7314} = 1{,}37$

Javob: $n = 1,37$

4-masala

Fokus oralig‘I 40 cm bo‘lgan ikki tomoni qavariq linzada buyumning haqiqiy tasviri ikki marta kattalashgan bo‘lishi uchun buyumni linzadan qanday uzoqlikda o‘rnatish kerak?

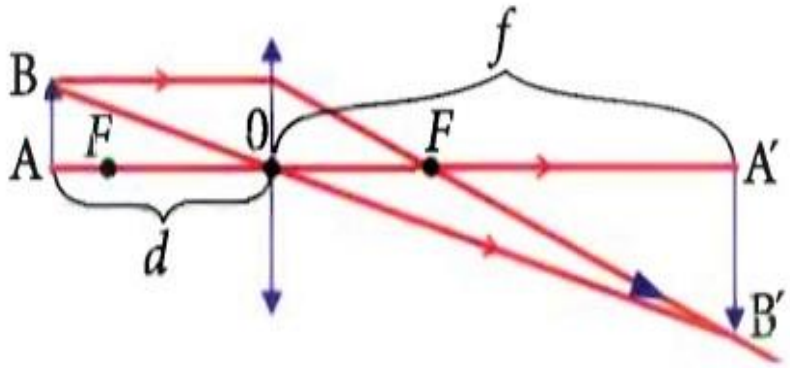
Berilgan:

$F = 40\ cm = 0{,}4\ m$

$K = 2$

Topish kerak: $d - ?$

Formula:



$$\frac{1}{F} = \frac{1}{d} + \frac{1}{f}\ ; \qquad K = \frac{f}{d}; \qquad f = Kd$$

$$\frac{1}{F} = \frac{1}{d} + \frac{1}{Kd} = \frac{1}{d}\left(1 + \frac{1}{K}\right)$$

$$d = F\left(1 + \frac{1}{K}\right)$$

Yechish: $d = 0{,}4\ m \cdot \left(1 + \frac{1}{2}\right) = 0{,}6\ \text{m} = 60\ \text{cm}$

Javob: $\boldsymbol{d = 60\ \text{cm}}$

5- masala

Ikki tomonlama qavariq linzaning egrilik radiuslari $R_1 = 20\ \text{cm}$ va $R_2 = 30\ \text{cm}$, sindirish ko‘rsatkichi 1,5.

Linzaning bosh fokus oralig‘ini toping.

Berilgan:

$R_1 = 20\ \text{cm} = 0{,}2\ \text{m}$

$R_2 = 30\ \text{cm} = 0{,}3\ \text{m}$

$n = 1{,}5$

Formula:

$$F = \frac{1}{(n-1)\cdot\left(\frac{1}{R_1}+\frac{1}{R_2}\right)}$$

Topish kerak: $F - ?$

Yechish: $$F = \frac{1}{(1{,}5-1)\cdot\left(\frac{1}{0{,}2\ m}+\frac{1}{0{,}3\ m}\right)} = 0{,}24\ m = 24\ cm$$

Javob: $\boldsymbol{F = 24\ cm}$

Mustaqil bajarish uchun topshiriqlar

1. Quyosh yorug‘ligi suv yuziga tushmoqda. Agar qaytish burchagi 30º bo‘lsa, tushish va sinish burchagi qanday?

2. Yorug‘likning yupqa dastasi havodan qandaydir bir suyuqlikka o‘tmoqda. Agar tushish burchagi 30º, sinish burchagi 15º bo‘lsa, suyuqlikning nur sindirish ko‘rsatkichini toping.

3. Fokus oralig‘i 40 cm bo‘lgan ikki tomoni qavariq linzada buyumning haqiqiy tasviri ikki marta kichraygan bo‘lishi uchun buyumni linzadan qanday uzoqlikda o‘rnatish kerak?